



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

CONCENTRATION CUMENE QUENCH DRUM (D-2301)

Docu. No. : 14780-8120-PS-D2301
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

CONCENTRATION CUMENE QUENCH DRUM

(D-2301)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-D2301
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 22, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 2, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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Revision
Indication



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VESSELS

PROJECT SPECIFICATION

963766 — 301

SHEET 3 / 4

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	29MAR12	JG	MYW				

General Notes

- A. Vessels shall be constructed in accordance with the following unless otherwise specified on the individual vessel specification sheet.

Standards:

UOP Standard Specification 3-11
UOP Standard Drawing 8-121

- B. Although the resulting inside diameter may be slightly different than specified, vessels without trays or packing, may be fabricated from pipe.
- C. The dimensioned location of some of the vessel level nozzles may be noted as min (minimum) or max (maximum). It is the responsibility of the contractor to adjust these dimensions to suit the specific level requirements specified on the UOP Piping and Instrument Diagram(s) and on UOP Standard Drawing 8-121. When additional nozzles are required for level instrumentation, the contractor shall specify them consistent with the size, class and facing requirements specified in the UOP Project Specification 301-Vessels and on the UOP Piping and Instrument Diagram(s).
- D. When system of measurement is metric, all dimensions are in millimeters unless noted otherwise.
- E. Contractor shall not locate TI thermowells in a downcomer. Contractor shall also ensure that TI thermowells are located to avoid interference with downcomers, weirs, distributors, collectors, baffles, chimneys, traps and all other vessel internals.

Other specifications
referenced in this
specification:

UOP Standard Specification: 3-11, 3-15

UOP Standard Drawing: 3-115, 3-122, 3-123, 3-183, 3-190, 3-272, 3-278, 3-280, 3-305, 3-313, 3-322, 8-121

UOP Project Specification: 963766-307, 963766-401, 963766-808



R
E
V

VESSELS

SERVICE	CONCENTRATION CUMENE QUENCH DRUM	ITEM NO	D-2301
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Design	INT	5 kg/cm ² (g)	120° C
Conditions	EXT	FULL VACUUM	195° C
at Top	Metal Temperature(Min)	15° C	
Operating	0.06 kg/cm ² (g)	38° C	
Conditions			° C

Radiograph	MR
Postweld Heat Treat	MR
Material	Corrosion Allowance mm (Min)
KILLED CARBON STEEL	3

This Vessel is in the Following Service(s):	
Hydrogen	NO
Hydrofluoric (HF) Acid	NO
Hydrofluoric (HF) Acid (with Resid. Elem. Control)	NO
Wet Hydrogen Sulfide (H ₂ S)	NO
Severe Cyclic	NO
Caustic	NO
Carbonate	NO
Amine	NO
Note:	

Vessel Heads	
Vessel	2:1 ELLIPTICAL

Accessories Applied by Fabricator	
Ladder & Platform Clips	AS REQUIRED
Insulation Clips & Rings	C (50) YES
Vessel Support	YES

Nozzles and Manways			
Mark	No	Size Inches	Service
AD	2	2	DRAIN
AI	1	2	INLET
AK	2	2	LT
AL	2	2	LG (2)
AM	1	24 ID	MANWAY
AO	1	10	OUTLET
AP	1	2	PI (2)
AX	1	2	INLET
AY	1	6	OUTLET
BB	1	8	VENTILATION
BD	2	2	DRAIN
BC	1	3	EQUALIZING
BM	1	24 ID	MANWAY
BN	1	8	OUTLET
BP	1	6	OUTLET
BT	1	2	TI (2)
BX	1	2	INLET
BY	1	2	INLET
BZ	1	3	OUTLET

Class	- ASME CL150
(2)	- ASME CL300
Facing	- RAISED FACE

Normal Liquid Level	= 840 mm ABOVE VESSEL CENTERLINE
Specific Gravity	= 0.84

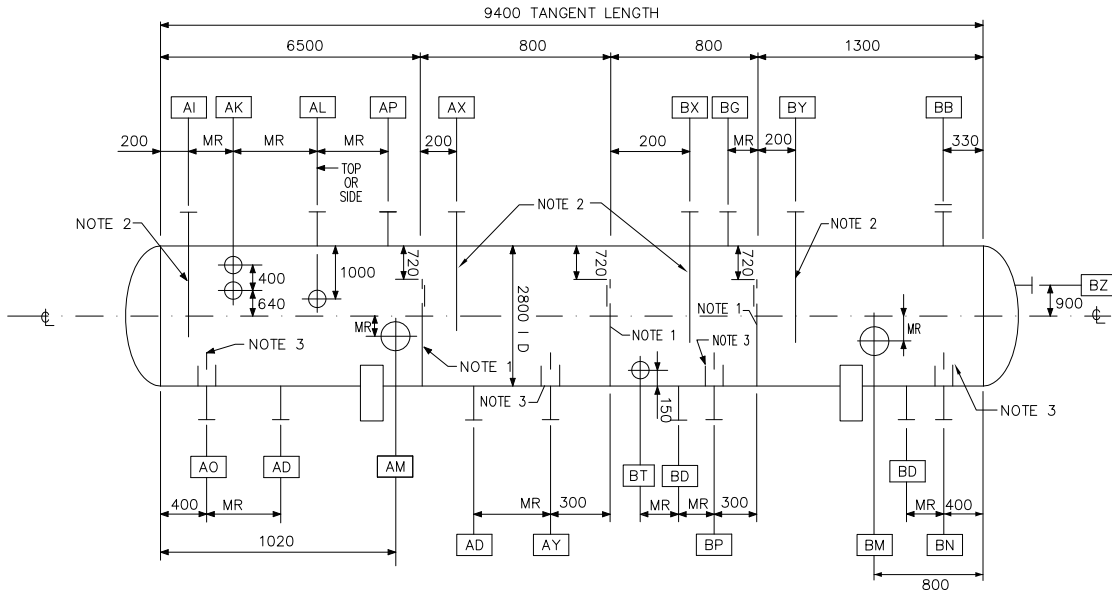
NOTES:

1. PROVIDE 10mm MIN THICK PARTITION, WITH STIFFENERS AS REQUIRED (WITH REMOVABLE GASKETED ACCESS SECTION).
2. PROVIDE DISTRIBUTOR UOP STD DWG 3-183, TYPE C.
3. PROVIDE 150 mm HIGH RAISED VORTEX BREAKER, UOP STD DWG 3-123.
4. THE CENTERLINE OF UPPER LEVEL NOZZLES LOCATED ON THE SIDE OF VESSEL MUST BE A DISTANCE FROM THE TOP OF THE SHELL EQUAL TO 5% OF VESSEL DIAMETER.

PROJECT SPECIFICATION

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REV	DATE	BY	APP'D	REV	DATE	BY	APP'D
0	29MAR12	JG	MYW				
1	13AUG12	TS	MYW				



Drawings Referred to in this Specification

3-123	3-183					
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